Total No. of Questions : 8]

SEAT No. :

P1452

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[6004]-574

B. E. (Information Technology)

DISTRIBUTED SYSTEMS

(2019 Pattern) (Semester - VIII) (414450)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Figures to right indicate full marks.*
3) *Neat diagrams must be drawn wherever necessary.*
4) *Assume suitable data, if necessary.*

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Q1) a) What are the general characteristics of inter-process communication? What are various types of Synchronous and asynchronous communication in IPC? Why it is blocking receive has no disadvantages in Java. **[8]**

b) What is Network virtualization? What are the types of overlay networks? What are the various advantages of overlay networks? **[9]**

OR

Q2) a) Describe Lamport's clock algorithm and give an example of how to order events in a distributed system. What are the algorithm's limitations? **[8]**

b) Why do we need an election algorithm? Using appropriate diagrams. describe the bully algorithm step-by-step. **[9]**

Q3) a) What are the key issues in Replica Management? Explain the following with respect to content replication and placement with suitable diagram. **[9]**

i) Permanent Replicas

ii) Server-Initiated Replicas

iii) Client-Initiated Replicas

b) What is a primary-based protocol in a consistency protocol? Explain the working of replicated write protocols with active replication. **[9]**

OR

P.T.O.

Q4) a) What is checkpointing in a distributed system? Explain the working of Coordinated checkpointing recovery mechanism. **[9]**

b) What are two primary reasons for replication? Explain the causal consistency model with suitable example using distributed shared database. **[9]**

Q5) a) Describe the architecture of Sun Network File System in details. **[8]**

b) What is a Directory Service? What is the difference between DNS and x500? Describe in detail the components of X.500 service architecture. **[9]**

OR

Q6) a) Describe the design of a peer-to-peer download system designed to support very large multimedia files. How does the BitTorrent protocol operate? **[8]**

b) What are web services? Describe with a suitable diagram the general organization of the Apache web server. **[9]**

Q7) a) Explain the following in brief: **[9]**

i) Wearable devices

ii) PVM

iii) JINI

b) Explain with an application (e.g., Travel Booking Service) various key components of Service Oriented Architecture. **[9]**

OR

Q8) a) Explain in brief the following Distributed System monitoring tools: **[9]**

i) Zabbix

ii) Nagios

b) Explain in brief following Microkernels. **[9]**

i) Mach

ii) CHORUS

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